

CityViews: Benchmarking Satellite and Street-Level Imagery in Urban Vision-Language Models

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Abstract. Pairing satellite and street-level imagery is widely assumed to improve urban vision-language models (VLMs), but aggregate benchmarks rarely reveal when the second view actually helps. We introduce CityViews, an OpenStreetMap-grounded benchmark designed to measure this marginal value directly. CityViews pairs marked satellite tiles with road-aligned street-level images across 40 cities and 12 tasks, spanning urban-attribute questions about a single location and cross-view questions whose answers require correspondence between views. Every question is evaluated under three matched input conditions – satellite only, street-level only, and both – from which we derive *fusion synergy*: the accuracy gain of both views over the stronger single-view input. Even when a fine-tuned 9B VLM is provided with the satellite tile and all four street-level views, accuracy on seven urban-attribute tasks improves by only +1.6 points over the stronger single view. On five cross-view tasks, by contrast, synergy reaches approximately +47 points for the same model, where single-views sit near chance. The small urban-attribute gain is robust across the raw backbone, inference-time pruning, and reduced training-time street-view budgets. CityViews shows that the value of a second urban view is task-dependent; large when the answer depends on the relationship between views, but small when either view already provides sufficient evidence.

Keywords: Geosensing · Vision-language models · Multi-view fusion

1 Introduction

A growing body of work treats satellite and street-level imagery as naturally complementary inputs for understanding cities; satellite tiles capture road layout, land use, and spatial connectivity, while street-level views reveal facades, signage, and local appearance that no aerial sensor can resolve [7, 29, 38]. The intuition is appealing, but it has rarely been put to a direct test. Most evaluations report one accuracy number over a heterogeneous pool of questions [7, 19, 33, 38], and that number conflates two very different situations: a question that one

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view could have answered alone, and a question that genuinely needs both. A model can score well on such a benchmark for either reason, and it remains unclear whether it has fused complementary evidence or simply relied on the stronger single view. Knowing the difference matters in practice. Street-level coverage is expensive to acquire, license, and serve at scale, so whether it provides measurable benefit beyond the satellite tile is a question worth answering before, not after, deployment. Throughout, we use *urban VLM* as shorthand for a general-purpose vision-language model applied or fine-tuned for such urban understanding tasks.

We address this gap with CityViews, an OpenStreetMap-grounded benchmark spanning 40 cities and 12 task families, each location is paired with one marked satellite tile and four road-aligned street-level views. Importantly, tasks are organized into two regimes by construction: urban-attribute questions (land use, road type, building height, transit density, and others) that concern a single location and may be answerable from either view alone, and cross-view questions (camera direction and satellite-street matching) that are answerable only by relating the two views to each other. Every question – identical text, identical answer choices – is posed three times: with the satellite tile alone, with the street views alone, and with both. The gap between the combined score and the better of the two single-view scores is what we call *fusion synergy*, and it is the quantity the rest of the paper measures. This protocol separates the two regimes cleanly within a single model – urban-attribute performance survives on either view alone, while cross-view performance exists only with both – and the split is not a training artifact: it persists for the unmodified backbone, under inference-time image pruning, and with fewer street-level views at fine-tuning.

Our contributions are threefold:

1. **CityViews, an OpenStreetMap-grounded benchmark** of 12 tasks across 40 cities, with matched satellite-only, street-view-only, and full-view evaluation modes built specifically to isolate the marginal value of a second imagery source.
2. **A measurement of fusion synergy that dissociates within a single model.** The same fine-tuned 9B VLM gains only +1.6 points from the combined input on urban-attribute tasks, but +46.6 points on cross-view tasks: whether the second view helps is a property of the task, not of the model.
3. **A view-budget robustness analysis.** The small urban-attribute gain is not an artifact of offering four street-level views: it persists when the model is trained with only the forward street view, and when street views are pruned at inference time.

2 Related Work

Remote-sensing VLMs rarely use street-level imagery. Recent instruction-tuned VLMs have substantially expanded remote-sensing image understanding, including GeoChat [14], SkySenseGPT [21], the LHRs-Bot family [17, 23], EarthGPT [36], and the multi-sensor generalist EarthDial [28]. These models

Table 1: CityViews compared with related urban and remote-sensing benchmarks. *Paired views* indicates coordinate-aligned overhead+street-level inputs; *matched ablation* indicates that the same questions are evaluated under satellite-only, street-view-only, and combined inputs.

Benchmark	Views	Task format	Labels	Matched ablation
RSVQA [20]	overhead only	VQA	OSM-derived	no
VRSBench [16]	overhead only	VQA/grounding	human-verified	no
UrBench [38]	paired	MCQ	mixed	no
CityLens [19]	paired	regression	socioeconomic	no
UrbanLLaVA [7]	sat., SV, OSM, traj.	VQA	mixed	no
CityCube [33]	paired	MCQ	geometric	no
CityViews (ours)	paired	MCQ	OSM-derived	yes

primarily operate on aerial or satellite imagery. When the term “multimodal” appears in this literature, it usually refers to multiple remote-sensing sensors, such as optical, SAR, or infrared imagery, rather than pairing satellite imagery with street-level views. The dominant evaluation suites follow the same pattern: RSVQA [20], VRSBench [16], and GeoChat’s benchmark pose questions over a single remote-sensing image and therefore do not measure the marginal value of adding a street-level view.

Multi-view urban benchmarks. Several recent benchmarks and models study urban understanding from paired or heterogeneous city-scale data. UrBench [38] evaluates coordinate-aligned satellite and street-level pairs and observes that multi-image and single-image performance can be close, but reports aggregate scores rather than a controlled input-mode ablation. CityLens [19], concurrent with our work, studies urban socioeconomic regression and finds street-level imagery competitive with the combined input, but focuses on a small set of regression targets and a single model. UrbanLLaVA [7] trains with satellite, street-view, OSM, and trajectory information, yet its attribute tasks are posed from satellite imagery alone. CityCube [33] instead targets cross-view spatial reasoning, a setting where correspondence between views is central. These works establish the importance of urban multi-view data, but they do not isolate when a second view provides limited marginal benefit and when cross-view correspondence is required. CityViews addresses this gap with a paired-view task taxonomy and a matched three-way ablation across satellite-only, street-view-only, and full-view settings. Table 1 situates CityViews among these benchmarks.

Cross-view geolocalization. Satellite-ground pairing is well established in cross-view geolocalization, where the goal is to match a ground-level image to an aerial or satellite reference [5, 34, 39]. In this setting, the two views are jointly necessary: methods succeed by modeling the distinct but linkable spatial structure shared across viewpoints [37]. Our cross-view tasks occupy this regime through camera-direction and satellite-street matching questions, for which single-view inputs are insufficient by construction.

The marginal value of a second modality. Beyond geosensing, prior work has shown that adding a modality does not automatically improve performance.

Wang *et al.* [31] show that the best single-modal network can outperform naive multimodal fusion, while Liang *et al.* [18] decompose multimodal interactions into redundancy, uniqueness, and synergy, showing that low-synergy regimes can arise. Conversely, multimodal learning can provably reduce achievable risk [12], and several systems that process satellite and street view images all together report gains from combining views [22,32]. Suel *et al.* [29], for example, show that satellite and street-level imagery improve regression of income, overcrowding, and environmental deprivation. Our goal is not to dispute such gains, but to make their conditions measurable for urban VLM evaluation.

3 CityViews Benchmark

CityViews is an OpenStreetMap [24]-grounded benchmark constructed for evaluating when satellite and street-level imagery should be combined in urban vision-language models. Each example is built around a geographic location and pairs a marked satellite tile with road-aligned street-level views. Each answer is derived from OSM tags or coordinate geometry rather than manual annotation. This limits the benchmark to properties that OSM and geometry can certify, but makes the labels reproducible: the same OSM snapshot and generation code regenerate the benchmark. The benchmark is designed around a matched input-mode comparison. For each question, the text and answer choices are held fixed while the visual input changes across three conditions: satellite only, street-view only, and both views. This makes it possible to measure the marginal value of the second view rather than reporting only aggregate performance with all images present.

3.1 Construction

Each *location* is sampled within one of 40 cities across 32 countries on six continents. A location is associated with three components: (i) one marked satellite tile, (ii) four street-level views captured at fixed road-aligned viewing directions (forward, backward, left, and right) and (iii) OSM-derived attribute labels. A seven-stage pipeline (Fig. 1) threads each sample through location sampling, satellite-tile retrieval, street-view capture, OSM enrichment, composite rendering, question generation, and validation/splitting.

The satellite tile is sourced per location by geography from sub-meter imagery: Esri World Imagery [6] as the global default, NAIP [30] in the US, and IGN [13] in France. We write “satellite tile” for the marked overhead tile, although most of these sub-meter sources are aerial orthoimagery. Esri imagery is typically 0.3 m/px to 0.5 m/px in cities, NAIP 0.6 m/px, and IGN 0.2 m/px; across the 34484-question corpus, 31489 questions (91%) use Esri, 1946 NAIP, and 1049 IGN. The satellite tile contains a marker indicating the queried location; this marker is part of the input and is held fixed across all input modes. Street-level views are retrieved from Google Street View [9] and retain Google’s original attribution; satellite imagery is © its respective provider (Esri, NAIP/USDA, IGN).

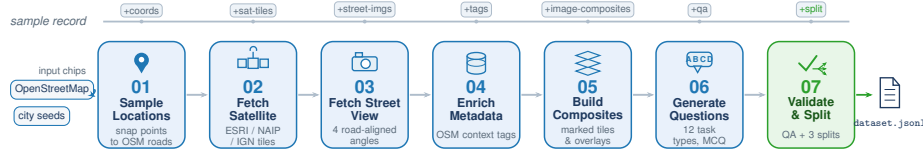


Fig. 1: The seven-stage generation pipeline. Every stage reads and enriches the same list of sample dictionaries; no human annotation is involved. Locations are sampled uniformly within each city’s bounding box and snapped to the road network, with the sampler seeded per city for reproducibility; “composites” are the 2×2 tilings that pack a location’s four street-level views (and, for mismatch questions, four candidate view sets) into single images.

3.2 Task Taxonomy

CityViews contains 12 different tasks in two groups (Fig. 2): seven urban-attribute tasks and five cross-view tasks.

Urban attributes. The urban-attribute group contains `land_use`, `building_height`, `urban_density`, `road_type`, `junction_type`, `amenity_richness`, and `transit_density`. Each task is a four-way multiple-choice question about the OSM context of one marked location. These tasks represent the setting in which either satellite or street-level imagery may already provide enough evidence, so the benchmark asks whether holding both views yields additional benefit.

Cross-view correspondence. The cross-view correspondence tasks are `camera_direction`, `mismatch_binary_{easy,hard}`, and `mismatch_mcq_{easy,hard}`. `camera_direction` asks which of four arrows on the satellite tile corresponds to the heading of a queried street view. The binary mismatch tasks ask whether a street-view set belongs to a satellite tile, and the multiple-choice mismatch tasks ask which of four street-view composites matches the tile. Easy variants draw distractors from another city, where vegetation, signage language, and vehicle types may provide shortcuts. Hard variants draw distractors from the same city, reducing these shortcuts and emphasizing geometric correspondence. These tasks require the relationship between the satellite and street-level views; a single view is insufficient by construction.

3.3 Label rules, options, and distractors

No model in the loop. Every question is instantiated from one of ten hand-written paraphrase templates per task; no LLM or VLM is involved at any stage of label derivation, question generation, or distractor construction. The pipeline’s only external calls fetch imagery and OSM data.

Label rules. Urban-attribute labels are computed from OSM tags and geometry inside a window matching the satellite-tile footprint (200×200 m for Esri and IGN tiles, 500×500 m for NAIP); `transit_density` instead counts deduplicated public-transport stops within a fixed 300 m radius. `land_use` follows a fixed priority cascade over `landuse` tags, dominant building types, and amenity ratios;

Every question is answered from **both** views:  **satellite tile** (overhead) +  **street view** (ground; 4 angles)

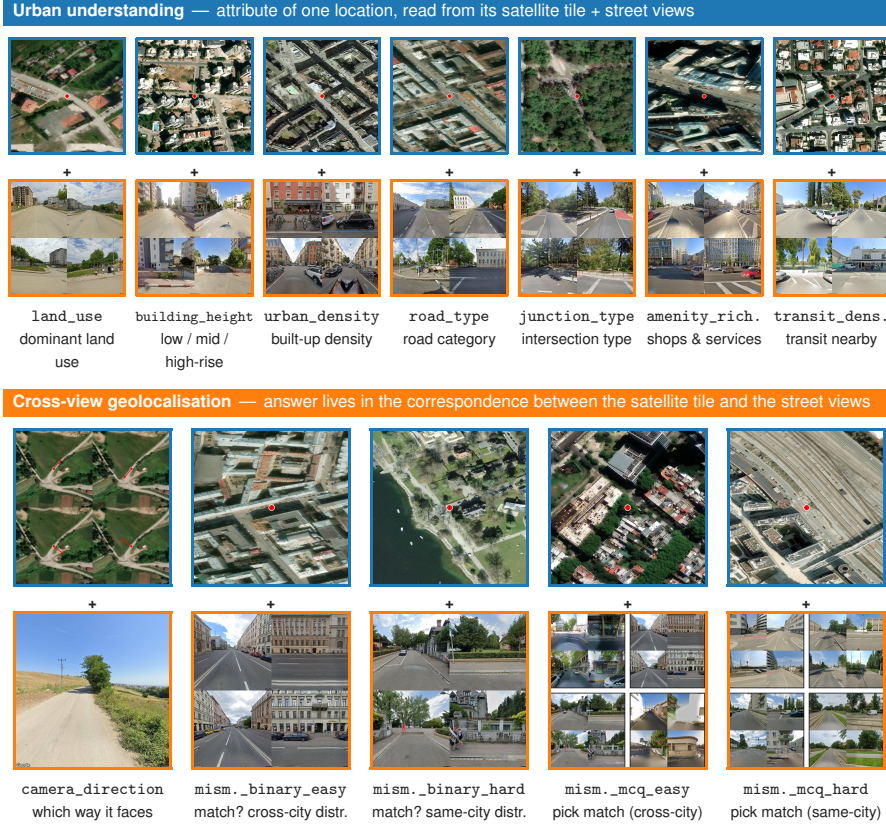


Fig. 2: The CityViews tasks. Each cell pairs a satellite tile (blue, top) with a street view (orange, bottom); every location provides one marked satellite tile and four road-aligned street views. **Top - seven urban-attribute tasks:** location attributes such as `land_use` and `building_height`, grounded in OSM context. **Bottom - five cross-view tasks:** `camera_direction` matches a heading arrow to a street-view query; `mismatch_binary/mcq` ask whether, or which, street-view set matches the tile, with *easy/hard* distractors drawn from another/the same city. (Imagery attribution: satellite tiles © Esri and its data providers; street-level imagery © Google.)

`road_type` is the OSM `highway` class of the road the location is snapped to; `junction_type` applies the priority roundabout → signalized → grade-separated → unsignalized. Numeric tasks use fixed bin edges: `building_height` by median `building:levels` (1–3, 4–7, 8–20, ≥21), `urban_density` by building count (≤15, 16–50, 51–150, >150), `amenity_richness` by amenity count (0, 1–4, 5–19, ≥20), and `transit_density` by stop count (0, 1–2, 3–5, ≥6).

Distractors and option order. Wrong answers are drawn deterministically from fixed per-task class inventories: the four-class tasks always present the full

Table 2: Record counts for the CityViews corpus: 34484 questions over 4168 locations in 40 cities. The two training strategies share one held-out benchmark split, which is defined first and kept identical across both.

Split set	Train	Validation	Benchmark
per_city	23,431	6319	—
seen_unseen	23,282	8179	—
benchmark (12 tasks)	—	—	4734

inventory, and `land_use` and `road_type` sample three of the remaining classes. For `camera_direction` the options are the four arrow renderings of the same tile; mismatch distractor street-view sets are sampled from other locations, cross-city for *easy* and same-city for *hard* variants. Option order is shuffled by an RNG seeded from the sample id and task name, so the correct-letter assignment is reproducible across regenerations; letter balance is monitored but not explicitly enforced. No class or per-city balancing is applied; class frequencies follow the OSM-tag distribution of the sampled locations.

3.4 Splits

Splitting is performed at the location level: all questions associated with a location remain in the same split within a given training strategy. We first define a held-out benchmark split identically for all strategies by selecting ten percent of the locations in each of the 40 cities. This produces 4734 benchmark questions covering all 12 tasks. The remaining locations are divided according to two training strategies (Table 2), which draw from the same pool but partition it differently, so their train/validation totals need not match. The *per-city* split sends 15% of each city’s remaining locations to validation, so every city appears in both training and validation; the *seen/unseen* split instead assigns all remaining locations of eight held-out cities (Istanbul, Moscow, Chicago, Seoul, Rio de Janeiro, Cape Town, Sydney, and Singapore) to validation, probing geographic generalization. Neither split touches the shared held-out benchmark, ensuring no data leakage. Locations with fewer than four street-level views are discarded; training splits are capped at 2000 questions per task, and validation and benchmark splits are never subsampled.

3.5 Evaluation protocol

We evaluate each model under a controlled image ablation. Every question is posed under three input conditions: **Sat**, containing only the marked satellite tile; **SV**, containing only the street-level view or views; and **Full**, containing both satellite and street-level imagery. The question text and answer choices are identical across the three conditions, so differences in performance are attributable to the supplied images rather than to language variation. The main fusion analysis uses the 2629 urban-attribute questions in the benchmark split; the 2105 cross-view questions provide the positive-control regime in which fusion should

be required. Per-task counts are uneven by design because they reflect OSM-tag availability rather than task importance: `building_height` has 190 items, `junction_type` has 334, and the other five urban-attribute tasks have 421 each.

4 Experiments

4.1 Setup

Our main fusion evaluation uses task-tuned 9B VLMs trained on CityViews. The headline model, CityViews-9B-SU, is Qwen3.5-9B [25] fine-tuned on the seen/unseen (SU) training split with LoRA [11]. We use rank 16, $\alpha = 16$, bfloat16 precision, a maximum image edge of 512 px, and adapt all linear layers, including those in the vision tower. Training uses an effective batch size of 48, learning rate 1.5×10^{-4} with cosine decay and 100 warmup steps, 8-bit AdamW with weight decay 0.01, completion-only loss on the answer letter, and early stopping on validation accuracy over a 9-epoch schedule. In the full condition, CityViews-9B-SU receives the marked satellite tile together with all four road-aligned street-level views, giving the model the maximum visual evidence available in the benchmark. We also train CityViews-9B-PC, an otherwise identical variant on the per-city (PC) split, to test whether the conclusions depend on the training split. These task-tuned models are used to measure fusion under controlled input modes: satellite only, street-view only, and both. Fine-tuning on the benchmark distribution makes this a favorable setting for fusion: the model has seen the task format and image arrangement during training, so small gains from the full input are less likely to be caused by prompt or task mismatch alone.

As zero-shot references we evaluate open-weight generalist VLMs spanning two model families and three scales: Qwen2.5-VL-7B [2], Qwen3.5 at 4B and 9B [25], and Gemma-4-12B [8]. These models are chosen because they are widely used open-weight backbones that accept interleaved multi-image inputs, which the three-way protocol requires; superscript ^r in the result tables denotes the unquantized raw checkpoint.

All evaluations use deterministic greedy decoding (temperature 0, at most 8 new tokens) with reasoning disabled, under a fixed prompt that lists the provided images, the question, and the lettered options and requests a single letter in reply. Answers are extracted with a letter-matching regular expression with an option-text fallback; malformed or refused outputs count as incorrect, with no retries. All models in the three-way fusion analysis run through a single evaluation harness, so within-model input-mode comparisons are exact. Fine-tuned models are evaluated at their 512 px training resolution and zero-shot baselines at up to 1120 px; for the zero-shot rows of Tab. 4, the four street-level views are packed into a single 2×2 composite rather than passed separately.

4.2 Three-Way Fusion Evaluation

We evaluate each question under three matched input conditions: **Sat**, **SV**, and **Full**. The resulting fusion synergy, $\text{Full} - \max(\text{Sat}, \text{SV})$, measures how much the

Table 3: Three-way fusion evaluation on the held-out benchmark. Sat, SV, and Full denote satellite-only, street-view-only, and combined inputs. We report $\Delta_{\text{sat}} = \text{Full} - \text{Sat}$, $\Delta_{\text{sv}} = \text{Full} - \text{SV}$, and Synergy = $\text{Full} - \max(\text{Sat}, \text{SV})$. Highlighted rows show per-task means; in particular, the highlighted Synergy is the unweighted mean of per-task synergies, not the synergy of the pooled accuracies (the caption of Fig. 3 reports the latter). Urban-attribute tasks yield small synergy, whereas cross-view tasks gain sharply when both views are provided. By construction these questions require relating the two viewpoints, so single-view accuracy stays near chance ($\sim 50\%$ binary, $\sim 25\%$ MCQ).

	Task	Sat	SV	Full	Δ_{sat}	Δ_{sv}	Synergy
CityV	ws-9B-SU						
	Urban-attribute tasks						
	amenity_richness	53.0	55.6	55.6	+2.6	+0.0	+0.0
	building_height	63.7	66.3	66.8	+3.2	+0.5	+0.5
	junction_type	74.0	71.3	76.3	+2.4	+5.1	+2.4
	land_use	72.0	72.5	73.9	+1.9	+1.4	+1.4
	road_type	69.6	76.7	76.7	+7.1	+0.0	+0.0
	transit_density	45.8	44.9	49.4	+3.6	+4.5	+3.6
	urban_density	71.5	64.8	74.6	+3.1	+9.7	+3.1
	Overall	64.2	64.6	67.6	+3.4	+3.0	+1.6
	Cross-view tasks						
	camera_direction	22.8	25.2	43.7	+20.9	+18.5	+18.5
	mismatch_binary_easy	51.8	52.7	98.8	+47.0	+46.1	+46.1
	mismatch_binary_hard	43.7	44.2	87.9	+44.2	+43.7	+43.7
	mismatch_mcq_easy	24.7	23.3	92.6	+67.9	+69.4	+67.9
	mismatch_mcq_hard	25.4	29.7	84.8	+59.4	+55.1	+55.1
	Overall	33.7	35.0	81.6	+47.9	+46.6	+46.6
CityV	ws-9B-PC						
	Urban-attribute tasks						
	amenity_richness	53.4	56.5	58.7	+5.2	+2.1	+2.1
	building_height	67.4	66.3	66.8	-0.5	+0.5	-0.5
	junction_type	73.4	68.6	75.1	+1.8	+6.6	+1.8
	land_use	75.5	75.1	75.3	-0.2	+0.2	-0.2
	road_type	72.5	77.4	76.5	+4.0	-0.9	-0.9
	transit_density	43.7	45.6	50.1	+6.4	+4.5	+4.5
	urban_density	61.8	62.7	65.1	+3.3	+2.4	+2.4
	Overall	63.9	64.6	66.8	+2.9	+2.2	+1.3
	Cross-view tasks						
	camera_direction	24.9	25.2	42.3	+17.3	+17.1	+17.1
	mismatch_binary_easy	51.1	52.7	97.4	+46.3	+44.7	+44.7
	mismatch_binary_hard	45.8	44.4	90.7	+44.9	+46.3	+44.9
	mismatch_mcq_easy	26.4	23.5	96.2	+69.8	+72.7	+69.8
	mismatch_mcq_hard	23.3	22.8	86.0	+62.7	+63.2	+62.7
	Overall	34.3	33.7	82.5	+48.2	+48.8	+47.8
Qwen3.5-9B	raw, zero-shot						
	Urban-attribute tasks						
	amenity_richness	36.3	33.2	35.9	-0.5	+2.6	-0.5
	building_height	48.4	59.5	62.6	+14.2	+3.2	+3.2
	junction_type	50.6	66.5	65.0	+14.4	-1.5	-1.5
	land_use	48.9	56.1	54.6	+5.7	-1.4	-1.4
	road_type	37.8	64.4	59.9	+22.1	-4.5	-4.5
	transit_density	29.4	30.6	30.2	+0.7	-0.5	-0.5
	urban_density	37.3	39.2	38.2	+1.0	-1.0	-1.0
	Overall	41.3	49.9	49.5	+8.2	-0.4	-0.9
	Cross-view tasks						
	camera_direction	27.3	25.2	26.1	-1.2	+0.9	-1.2
	mismatch_binary_easy	49.6	53.0	80.3	+30.6	+27.3	+27.3
	mismatch_binary_hard	52.0	48.5	70.5	+18.5	+22.1	+18.5
	mismatch_mcq_easy	25.4	24.5	66.0	+40.6	+41.6	+40.6
	mismatch_mcq_hard	23.0	20.7	52.5	+29.5	+31.8	+29.5
	Overall	35.5	34.4	59.1	+23.6	+24.7	+22.9

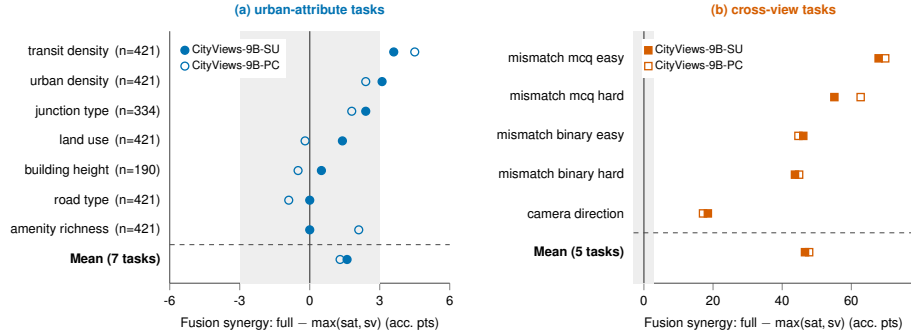


Fig. 3: Fusion synergy is task-dependent. Per-task gain of the full input over the better single view for both released models (filled: CityViews-9B-SU; open: CityViews-9B-PC), with the group mean as a summary row; values from Tab. 3. Urban-attribute tasks stay within a few points of the ± 3 -point equivalence band (defined in Sec. 4.2); the cross-view tasks, which require correspondence between views, show large gains. Pooled accuracies (Sat/SV/Full) over the 2629 urban-attribute questions: SU 64.2/64.6/67.6 (pooled synergy +3.0), PC 63.9/64.6/66.8 (+2.2); over all 4734 questions: SU 51.5/52.3/73.4, PC 51.6/51.7/73.4.

combined input improves over the stronger single-view input. Following the logic of equivalence testing [15], we fix ± 3 points as the smallest synergy of practical interest: per-task gains inside this band are read as no meaningful benefit from the second view. Table 3 reports this quantity per task; Fig. 3 summarizes the same pattern for both released models and reports pooled input-mode accuracies in its caption; pooled synergy (+3.0 for SU on urban tasks) differs from the headline +1.6, which is the mean of per-task synergies.

The main result is a strong within-model separation between the two task regimes. On the seven urban-attribute tasks, the main CityViews-9B-SU model gains only +1.6 points from the full input over the better single view. Per-task synergy remains small, never exceeding four points. In contrast, on the five cross-view tasks, the same model and the same evaluation protocol yield +46.6 points of synergy, raising performance from near-chance single-view accuracy (33.7/35.0% for Sat/SV) to 81.6% with both views. The urban synergy lies within the ± 3 -point equivalence band and is far separated from the cross-view synergy.

This separation shows why aggregate full-input accuracy is insufficient. For urban attributes, the second view adds little beyond the better single view, suggesting that the two image sources often provide redundant answer-relevant evidence for these tasks. For cross-view questions, however, the answer depends on the relationship between the satellite tile and the street-level views, and the full input is therefore decisive.

4.3 Zero-shot comparisons

Table 4 reports per-task accuracy under the full input setting, where the model receives the marked satellite tile together with all available street-level views.

Table 4: Per-task accuracy under the full input setting, where models receive the satellite tile and all available street-level views. Zero-shot VLMs are compared with CityViews-tuned variants. Superscript ^r denotes the unquantized raw model. Best result in each row is shown in **bold**; n is the number of examples, and *Overall* is the unweighted mean over the 12 different tasks.

Task	n	Qwen2.5 VL-7B	Qwen3.5 4B	Qwen3.5 9B	Qwen3.5 9B ^r	Gemma-4 12B	CityViews 9B-SU	CityViews 9B-PC
<i>Urban-attribute tasks</i>								
amenity_richness	421	31.8	35.4	36.3	35.9	35.6	55.6	58.7
building_height	190	57.9	61.6	56.8	62.6	61.6	66.8	66.8
junction_type	334	57.5	50.0	55.7	65.0	66.5	76.3	75.1
land_use	421	55.6	55.6	53.2	54.6	47.3	73.9	75.3
road_type	421	50.6	57.0	53.0	59.9	65.3	76.7	76.5
transit_density	421	28.0	34.2	31.4	30.2	28.0	49.4	50.1
urban_density	421	41.8	36.3	39.0	38.2	37.3	74.6	65.1
<i>Cross-view tasks</i>								
camera_direction	421	24.7	21.4	23.5	26.1	25.2	43.7	42.3
mismatch_binary_easy	421	59.1	74.4	87.2	80.3	79.8	98.8	97.4
mismatch_binary_hard	421	45.6	58.0	72.9	70.5	64.6	87.9	90.7
mismatch_mcq_easy	421	39.7	45.8	51.3	66.0	43.2	92.6	96.2
mismatch_mcq_hard	421	33.7	38.5	39.4	52.5	34.7	84.8	86.0
Overall	4734	43.8	47.3	50.0	53.5	49.1	73.4	73.4

This evaluation serves two purposes beyond the fusion analysis: it checks whether CityViews separates models under the richest input condition, and it shows how difficulty varies across the considered 12 different tasks. The task-tuned CityViews models perform best on all tasks, reaching 73.4% overall accuracy compared with 53.5% for the strongest zero-shot model. This indicates that generic VLM capability alone is not sufficient for strong performance on CityViews; adapting to the benchmark distribution provides a substantial advantage. At the same time, the benchmark does not saturate after fine-tuning. Performance varies widely across tasks: `amenity_richness` remains below 60%, while `mismatch_binary_easy` approaches 99%. The zero-shot models also show a broad range of overall performance, from 43.8% to 53.5%.

Table 5 evaluates the off-the-shelf remote-sensing VLMs on the satellite-only version of the seven urban-attribute tasks. These models provide external context rather than fusion baselines, since none can ingest street-level imagery. Their performance remains well below our task-tuned satellite-only reference. LHRs-Bot-Nova and EarthDial train on OSM-derived text and are marked as upper-bound references (*). This caveat does not affect the main fusion result, which is a within-model comparison across input modes where view-independent priors cancel.

4.4 Robustness

Small urban synergy is not specific to fine-tuning. We first repeat the same three-way input evaluation on the raw, zero-shot Qwen3.5-9B backbone. We observe a similar pattern in the results: mean synergy on the urban-attribute

Table 5: Zero-shot remote-sensing VLMs evaluated on the satellite-only version of the seven urban-attribute tasks. LHRs-Bot-Nova and EarthDial use OSM-derived training data and are marked as possible upper-bound references (*) due to training-data overlap with our OSM-grounded labels. None of these models ingests street-level imagery. GeoChat-7B and SkySenseGPT-7B build on Vicuna-1.5-7B [4] with CLIP ViT-L/14 [27] (504/336 px input); LHRs-Bot-Nova on Llama-3-8B [10] with SigLIP-L/14 [35] (336 px); EarthDial-4B-RGB on Phi-3-Mini [1] with tiled InternViT-300M [3].

Model	land _use	building _height	urban _density	junction _type	amenity _richness	road _type	transit _density	Overall
GeoChat-7B	48.7	30.5	13.5	8.4	13.5	10.7	21.4	21.0
SkySenseGPT-7B	53.4	34.7	14.5	30.8	12.6	18.1	26.1	27.2
LHRs-Bot-Nova*	46.3	28.9	17.8	35.3	17.8	26.1	34.2	29.5
EarthDial-4B-RGB*	47.5	32.6	30.4	52.7	25.4	46.1	29.7	37.8
CityViews-9B-SU	72.0	63.7	71.5	74.0	53.0	69.6	45.8	64.2

tasks is -0.9 points (Table 3, bottom block), while the cross-view tasks still benefit from both views. For example, `mismatch_mcq_easy` rises from 25.4% with the satellite tile alone to 66.0% with both views. The raw model is weaker overall and uses the two inputs differently: street-level imagery accounts for most of its urban performance gain ($\Delta_{\text{sat}}=+8.2$ versus $\Delta_{\text{sv}}=-0.4$), whereas the task-tuned model receives more balanced contributions from the two views. The shared observation is the near-zero marginal gain of the combined input on urban-attribute tasks, so the small urban synergy is not merely an artifact of LoRA fine-tuning. We run this full three-way evaluation only for the raw Qwen3.5-9B backbone; the other zero-shot models in Table 4 are evaluated only in the full setting.

A single forward street view largely preserves urban performance.

We next test whether the small urban synergy depends on giving the model all four street-level views. Table 6 compares the main five-image setting, which uses the satellite tile and four street-level views, with a two-image variant that uses only the satellite tile and the forward street-level view. (Because A and B are separately fine-tuned models, all columns, including `Sat`, differ between them.) On the held-out benchmark, using only the forward street-level view reduces full-input urban accuracy modestly, from 67.6% to 65.3%, while fusion synergy decreases from $+1.6$ to $+0.5$ points. The same pattern appears on the seen/unseen validation split, where both models show near-zero mean synergy.

We observe the same conclusion from an inference-time street-view pruning experiment with a stronger raw model, Qwen3.6-27B [26] (Table 7). Among urban-attribute questions answered correctly with the full street-view set, retaining only the forward street-level view preserves 86.9% of full-input correct predictions, close to the best single-view selector (87.5%) and above the worst single-view selector (81.6%). Removing street-level imagery entirely still retains 61.7%, using only the satellite tile and question text. Together, the training-time and inference-time results indicate that, for urban-attribute tasks, the remaining three street-level views provide limited additional benefit beyond the forward view.

Table 6: Ablation on the number of street-view images. Model A uses the marked satellite tile and four street-level views, while Model B uses the marked satellite tile and only the forward street-level view. Both are Qwen3.5-9B LoRA models finetuned on the seen/unseen split and differ only in the street-view inputs. The right-most block reports the change from the five-image model to the two-image model, $\Delta = \text{Model B} - \text{Model A}$. $\text{Syn} = \text{Full} - \max(\text{Sat}, \text{SV})$. Means are computed over the seven urban-attribute tasks; for the validation split we report the seven-task mean only.

Task	n	Model A				Model B				Δ		
		Sat	SV	Full	Syn	Sat	SV	Full	Syn	Sat	Full	Syn
<i>Seen/unseen validation split</i>												
Overall	4484	60.7	62.2	62.9	+0.4	60.2	60.8	61.5	-0.1	-0.5	-1.4	-0.5
<i>Shared held-out benchmark</i>												
amenity_richness	421	53.0	55.6	55.6	+0.0	55.3	47.7	53.0	-2.4	+2.4	-2.6	-2.4
building_height	190	63.7	66.3	66.8	+0.5	60.0	62.6	63.7	+1.1	-3.7	-3.2	+0.5
junction_type	334	74.0	71.3	76.3	+2.4	71.6	66.5	74.0	+2.4	-2.4	-2.4	+0.0
land_use	421	72.0	72.5	73.9	+1.4	72.5	71.7	72.9	+0.5	+0.5	-0.9	-0.9
road_type	421	69.6	76.7	76.7	+0.0	72.7	76.0	74.6	-1.4	+3.1	-2.1	-1.4
transit_density	421	45.8	44.9	49.4	+3.6	46.3	46.1	49.2	+2.9	+0.5	-0.2	-0.7
urban_density	421	71.5	64.8	74.6	+3.1	69.6	59.1	70.1	+0.5	-1.9	-4.5	-2.6
Overall	2629	64.2	64.6	67.6	+1.6	64.0	61.4	65.3	+0.5	-0.2	-2.3	-1.1

Table 7: Inference-time street-view pruning with Qwen3.6-27B. We report the retention rate (%) of full-input correct predictions after pruning the street-view set to k images, evaluated over the seven urban-attribute tasks. *Forward* keeps only the road-aligned forward view; *Top/Bottom* keep the best/worst selected subsets of size k ; *Random* is seed-averaged. *Zero* removes all street-level views while retaining the satellite tile. A single forward view retains most full-input correct predictions, close to the best single-view selector.

k images	Forward	Top	Random	Bottom	Zero
$k=0$ (no SV)	—	—	—	—	61.7
$k=1$	86.9	87.5	83.0	81.6	—
$k=2$	—	91.2	89.0	87.0	—
$k=3$	—	93.7	91.9	89.8	—

5 Discussion

CityViews shows that satellite-street fusion is task-dependent rather than uniformly beneficial: the combined input improves little over the stronger single view on urban-attribute questions but gains sharply on cross-view questions, so the second imagery source is most valuable when the task requires correspondence between views, not merely additional visual context. This should be read as a behavioral statement about current VLMs under our protocol. Our result refines, rather than contradicts, prior reports that combining satellite and street-level imagery can help. Suel *et al.* [29], for example, improve regression of income and deprivation by fusing the two views. We study multiple-choice urban-attribute recognition with different labels, models, and a controlled

input-mode ablation. These regimes can coexist: fusion may help for some urban prediction tasks while adding little for others.

Limitations and Scope. Our claim is limited to CityViews: urban-attribute tasks posed as four-option multiple-choice questions at the satellite-tile resolutions used in our benchmark. We do not claim that satellite imagery is uninformative or that fusion fails in general. Fusion synergy should be interpreted behaviorally: it measures the gain a model extracts from both views, not the information-theoretic complementarity of the views. A small value may mean that the views contain redundant answer-relevant evidence, or that the model fails to combine complementary cues. The strong cross-view gains make pure inability unlikely, but do not fully separate these explanations. CityViews is also constrained by its data sources: coverage is Europe-weighted, OSM-tag sparsity leads to uneven task counts, and labels certify OSM *context* rather than independently validated visual evidence. Relatedly, models pretrained on OSM-derived text may hold non-visual priors over our labels; the within-model comparison cancels view-independent priors, but absolute accuracies should be read with this in mind. For tasks such as amenity and transit density, this context may partly reflect nearby off-frame map attributes, limiting what any image input can recover. Future work could test finer satellite resolution, alternative labels, open-ended questions, and richer use of the four street-level views.

6 Conclusion

CityViews asks when a second urban view actually helps a VLM, and the answer is task-dependent. Given the satellite tile and all four street-level views, a fine-tuned 9B model gains little on urban-attribute questions, where either view alone often suffices, but gains sharply on cross-view questions, which require relating the two views. The pattern persists for the raw backbone, under inference-time view pruning, and with smaller training-time view budgets. CityViews is built deterministically from OpenStreetMap and its imagery providers, so it can be regenerated from source, and its matched three-way protocol gives future paired-view systems a direct test of whether a second imagery source pays for itself.

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